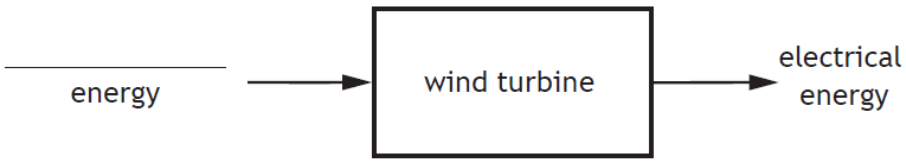
N5 Engineering Science – Systems Worksheet

Past Paper Questions – 2024, Q5, b

An incomplete diagram showing the **energy** transformation in a wind turbine is shown below.

(b) Complete the diagram by adding the missing input **energy**. **1**



The wind turbine uses feedback in its operation.

(c) State the type of control that uses feedback. **1**

Past Paper Questions – 2024, Q11

MARKS | DO NOT WRITE IN THIS MARGIN

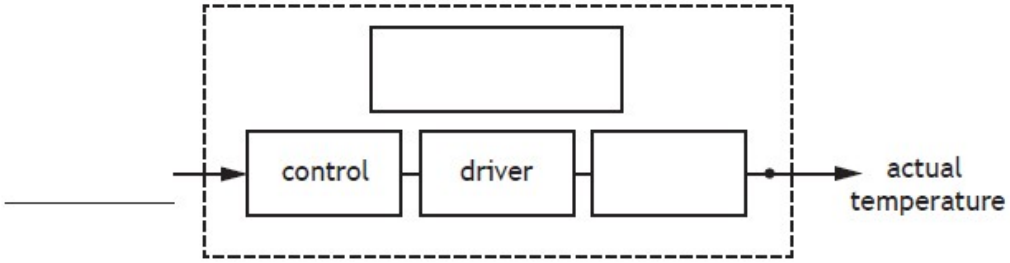
11. A dishwasher in a restaurant kitchen is shown.



Part of the specification for the dishwasher is given below:

- A user selects a set temperature.
- A sensor detects the actual water temperature and sends a signal to the control.
- The control compares the actual water temperature with the set temperature.
- If the actual water temperature is lower than the set temperature a heating element will switch on.

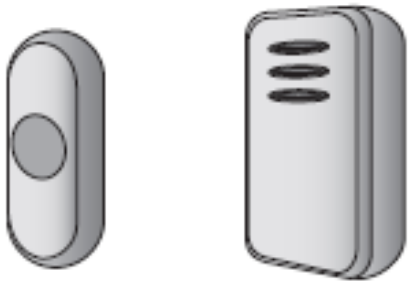
An incomplete sub-system diagram for the dishwasher is shown below.



(a) Complete the sub-system diagram with reference to the specification. 4

Past Paper Questions – 2023, Q1

1. The parts of a doorbell, consisting of a button and speaker, are shown.



button speaker

(a) Complete the system diagram shown below for the doorbell by adding the missing output. **1**



The doorbell operates without the use of feedback.

(b) State the type of control that uses feedback. **1**

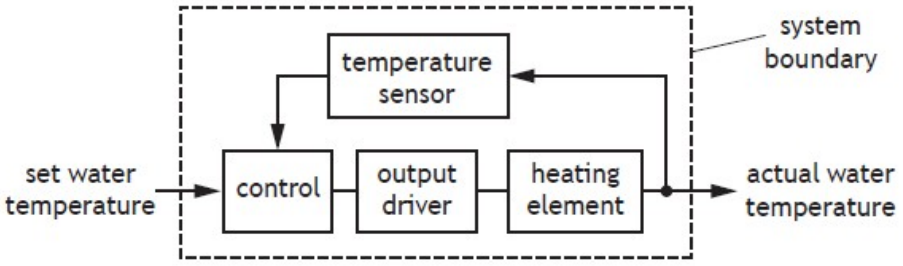
Past Paper Questions – 2023, Q12

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12. A washing machine is shown.



A sub-system diagram showing the control of the water temperature in the washing machine is shown below.



(a) Describe the reason for including a system boundary in a sub-system diagram. 1

(b) Describe, with reference to the sub-system diagram above, the control of the water temperature. 3

The water temperature is set . . .

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Past Paper Questions – 2022, Q2

2. An incomplete system diagram for a hair dryer is shown.



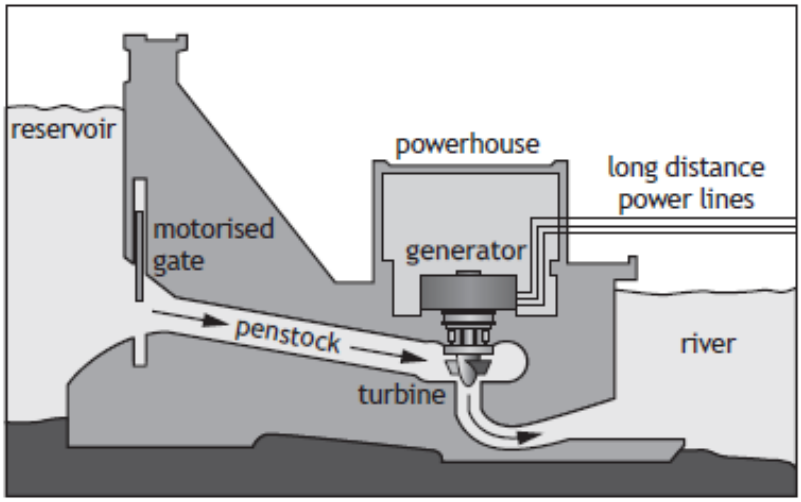
(a) Complete the system diagram above by adding the missing output. 1

(b) State the type of control that has **no** feedback. 1

Past Paper Questions – 2022, Q14

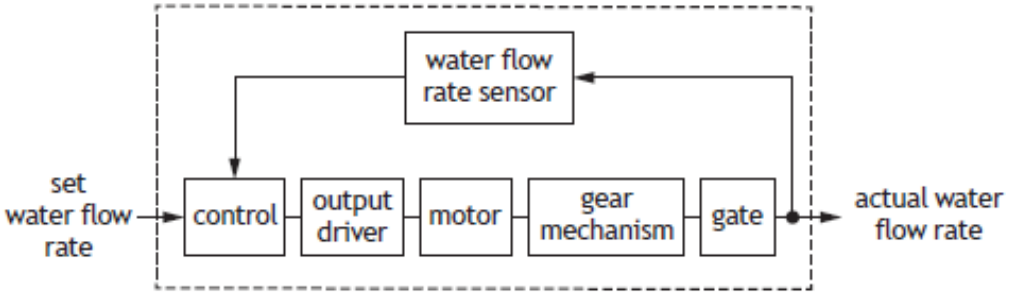
14. Hydropower is used to produce electricity from stored water in a reservoir. A diagram of a hydropower station is shown.

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A steady flow rate of water must move through the penstock to rotate the turbine. The water flow rate is adjusted by a motorised gate which can move up or down.

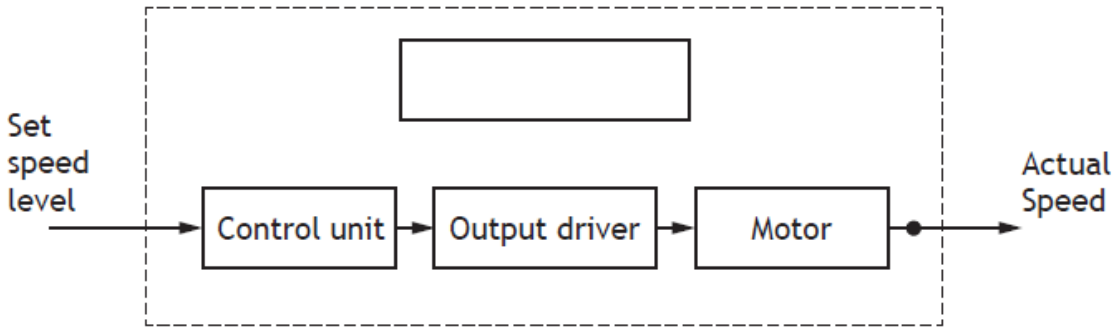
A sub-system diagram for the control of the motorised gate is shown.



(a) Describe (with reference to the sub-system diagram) the operation of the system. 5

Past Paper Questions – 2044, Q1

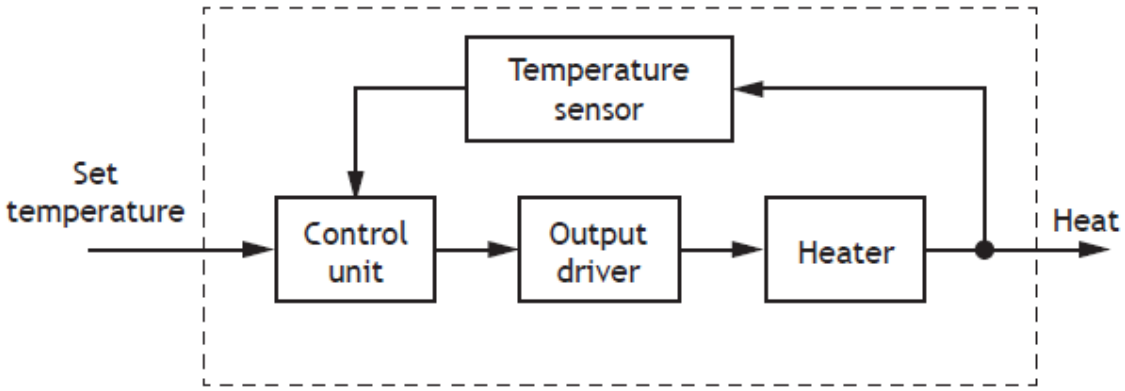
1. A closed loop system is used to control the speed of a motor.
   Complete the sub-system diagram shown below. 2



Past Paper Questions – 2015, Q3

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3. A diagram for a heating system is shown below.



(a) State the name of this type of diagram. 1

_____

(b) State the function of the output driver. 1

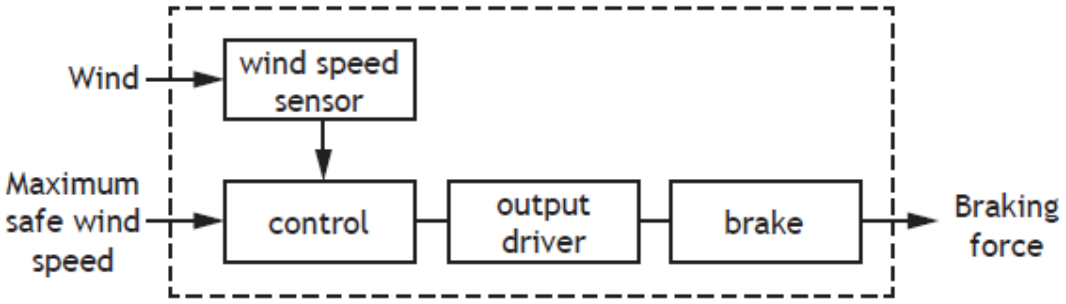
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Past Paper Questions – 2015, Q10b

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10. (continued)

A wind turbine can only be used when the wind speed is below a maximum safe level. A diagram for part of the braking system is shown below.



(b) Describe, using appropriate terminology, the operation of the wind turbine braking system. 3